Chapter 10

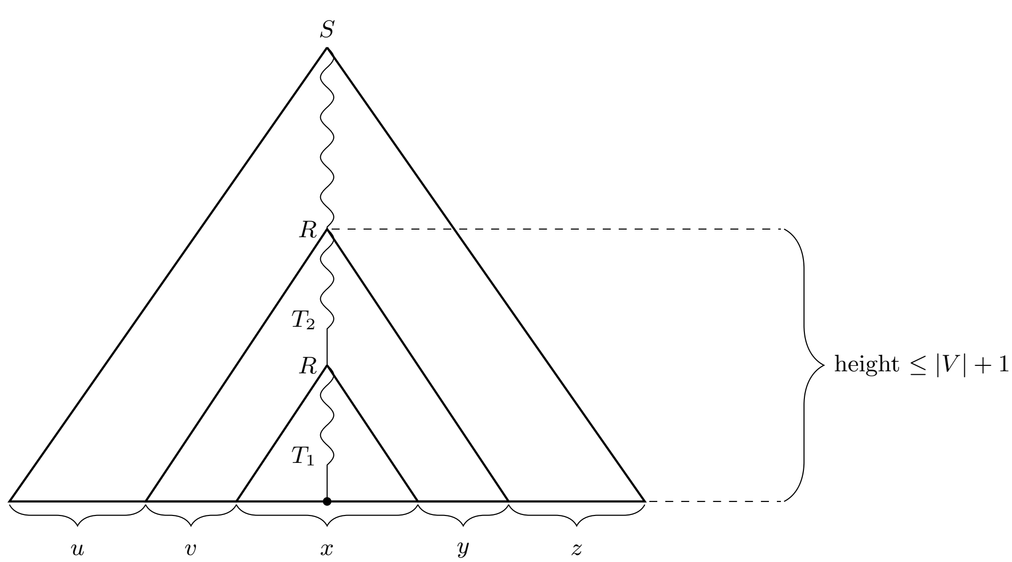
Non Context-free Languages

10.1 Pumping Lemma for Context-free Languages

We will prove a pumping lemma for context-free languages. Let L be a CFL and $G = (V, \Sigma, P, S)$ be a CFG such that $L = L(G)$. Let w be a string in L. Consider a smallest parse tree of w with respect to G (say $T_{G,w}$). Few observations:

- A path from the root to a leaf in $T_{G,w}$ is a sequence of variables ending with a terminal/ϵ.
- The height of a tree is the maximum number of edges on a path from the root to a leaf node.
- Let d be the maximum degree of a node in $T_{G,w}$. If the height of the tree is h, then $|w| \leq d^h$.
- Recall that w is the concatenation of the terminal symbols at the leaves of $T_{G,w}$, from left to right.

If $|w| \geq d^{|V|+1}$, then height of $T_{G,w}$ is at least $|V| + 1$ (no. of nodes is at least $|V| + 2$) and there exists a path in $T_{G,w}$ from root to a leaf on which it has at least $|V| + 1$ variables. Consider the lowest $|V| + 1$ variables on that path. By pigeon hole principle there exists a variable R which appears twice on that portion of the path. We define a partition of $w = uvxyz$ as illustrated in the figure below.



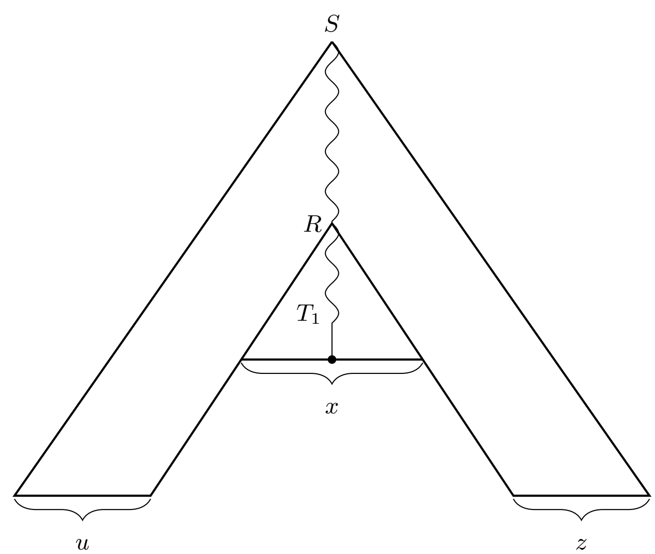
In the above parse tree for w, T_1 is the subtree rooted at the bottom R and it generates the string x and T_2 is the subtree rooted at the top R and it generates the string vxy.

Observation 16. *Suppose there are two internal nodes in a parse tree labelled with the same variable say A, and say T_1^A and T_2^A are the subtrees rooted at these two nodes respectively. If we replace T_1^A with T_2^A or vice versa then we will still get a parse tree for* some *string in the language of the grammar (essentially the string formed by concatenating the leaves from left to right).*

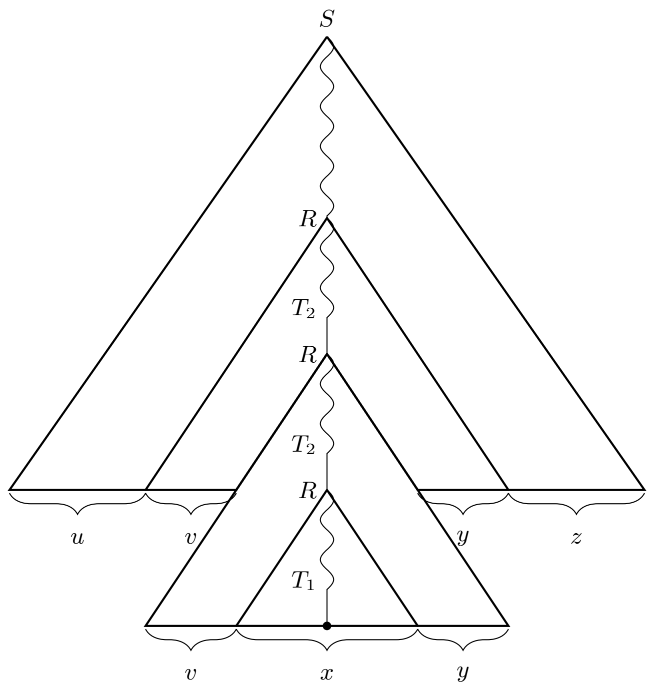
- Since height of T_2 is at most $|V| + 1$, therefore $|vxy| \leq d^{|V|+1}$.

- Moreover since $T_{G,w}$ is the smallest parse tree of w with respect to G, therefore T_1 cannot be substituted for T_2 to get the same string w. This implies that both v and y cannot be the empty string. Therefore $|vy| > 0$.

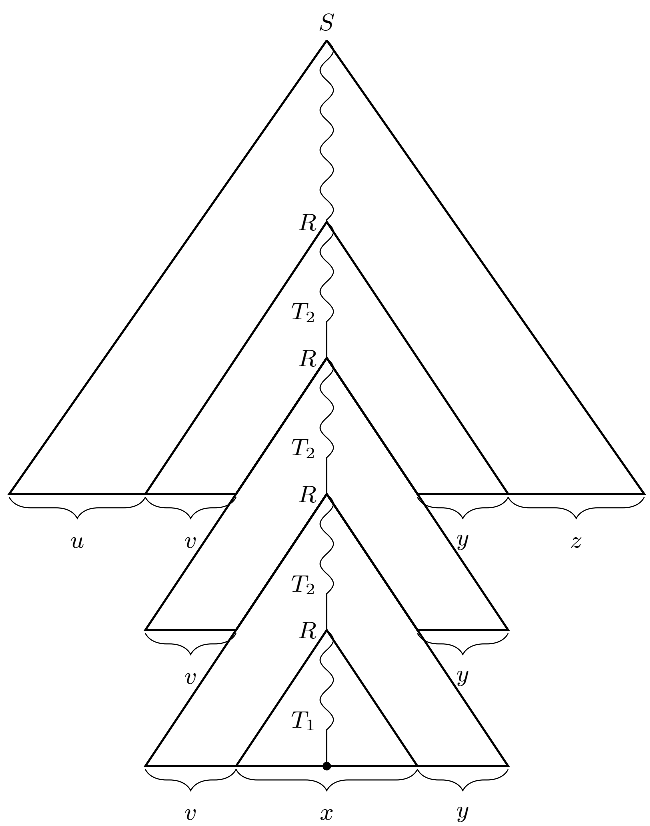
By Obversation 16 if we replace T_2 with T_1 we get the parse tree of the string uxz and hence this string is in L. The parse tree is shown below.



Similarly if we replace the T_1 with T_2 we get the parse tree of the string uv^2xy^2z and hence this string is in L. The parse tree is shown below.



Once again if we replace the T_1 with T_2 in the above parse tree we get the parse tree of the string uv^3xy^3z and hence this string is in L as well. The parse tree is shown below.



We can generalize and extend the above argument to show that for all $i \geq 0$, $uv^ixy^iz \in L$. Now by setting $p = d^{|V|+1}$ we get the following theorem.

Theorem 17 (Pumping Lemma for Context-free Languages)**.** *Let L be a context-free language. Then there exists an integer $p > 0$, such that for all $w \in L$ of length at least p, there exists a partition of $w = uvxyz$ such that $|vxy| \leq p$, $|vy| > 0$, and for all $i \geq 0$, $uv^ixy^iz \in L$.*

Remark. The choice of p for a CFL L is solely dependent on the CFG that we choose for L. Recall that $p = d^{|V|+1}$. Here d is the maximum number of symbols in the right hand side of a substitution rule in the CFG and V is of course the variable set of the CFG. Hence a different grammar for the same language might give a different p.

To prove that languages are not context-free, the pumping lemma will be used in its contrapositive form.

Theorem 18 (Contrapositive form of Pumping Lemma for CFLs)**.** *Let L be a language. If*

- $\forall p \geq 0$, *(opponent's move)*
- $\exists w \in L$ *with* $|w| \geq p$, *such that,* *(your move)*
- $\forall$ *possible partitions of* w *as* $w = uvxyz$, *satisfying* *(opponent's move)*
  - $|vxy| \leq p$, *and*
  - $|vy| > 0$,
- $\exists i \geq 0$ *such that* $uv^ixy^iz \notin L$, *(your move)*

then L is not context-free.

10.2 Examples of Non Context-free Languages

1. $$L_1 = \{a^nb^nc^n \mid n \geq 0\}$$

   Given p, choose $w = a^pb^pc^p$. Now for any partition $w = uvxyz$, set $i = 2$. We show below that $w' = uv^2xy^2z$ is not in L_1.

   Consider the string vxy. Since $|vxy| \leq p$, therefore vxy cannot contain all three symbols. More specifically, it does not contain either a or c. Assume that it does not contain c's. Also since v and y cannot both be empty, therefore w' will have more number of either a's or b's than the number of c's. Hence $w' \notin L_1$. The case when w' does not contain a's is analogous.

2. $$L_2 = \{ww \mid w \in \{a, b\}^*\}$$

   Given p, choose $w = a^pb^pa^pb^p$. Clearly $w \in L_2$ and has length at least p. Now for any partition $w = uvxyz$, consider the following cases.

   Case 1: vxy has only a's or only b's. We set $i = 2$ and let $w' = uv^2xy^2z$. Assume vxy lies in the first block of a's. Let $|vy| = k$. Now $0 < k \leq p$. As a result the first half of w' is $a^{p+k}b^{p-k/2}$ and second half of w' is $b^{k/2}a^pb^p$. Clearly the strings are not equal and hence $w' \notin L_2$.

   If vxy lies in any other block, the argument is analogous.

Case 2: vxy has both a's and b's. We set $i = 0$ and let $w' = uxz$. Assume vxy straddles the first boundary between a's and b's. Let $vy = a^{k_1}b^{k_2}$. Note that Now $0 < k_1 + k_2 \leq p$. Then $w' = a^{p-k_1}b^{p-k_2}a^pb^p$. Then the first half of w' is $a^{p-k_1}b^{p-k_2}a^{\frac{k_1+k_2}{2}}$ and the second half is $a^{p-\frac{k_1+k_2}{2}}b^p$. Clearly the strings are not equal and hence $w' \notin L_2$.

If vxy straddles any other boundary, the argument is analogous.

Remark. Note that in the above proof we could have fixed $i = 0$ or $i = 2$ for both the cases. But that would make the argument a little more tedious. Also the above proof illustrates the fact that i can vary on a case by case basis.

Exercise 20. Prove that the following languages are not context-free.

(a) $L_1 = \{a^nb^mc^nd^m \mid n, m \geq 0\}$

(b) $L_2 = \{0^n1^{n^2} \mid n \geq 0\}$

(c) $L_3 = \{0^n \mid n \text{ is prime}\}$

Chapter 11

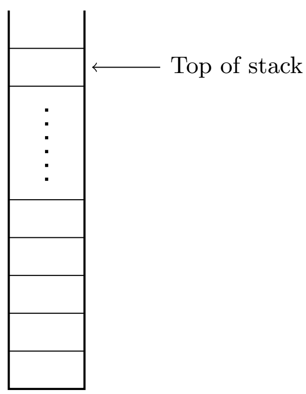
Pushdown Automaton

11.1 Pushdown Automata

It is an ϵ-NFA appended with a stack.

11.1.1 Reviewing a Stack

A stack is a data structure that allows addition/deletion/access of an element only at the top of a stack.



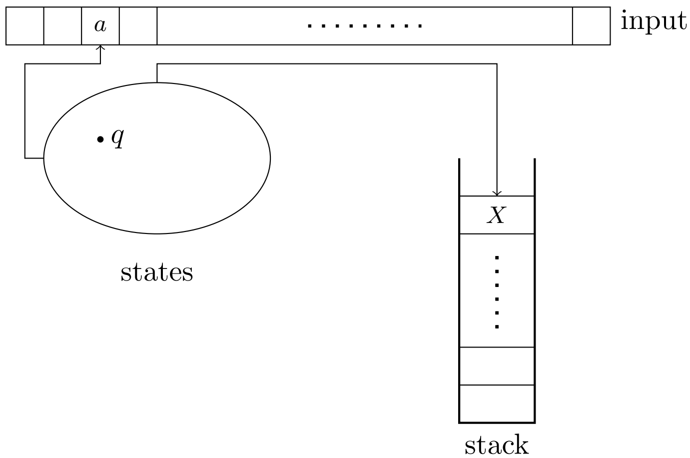
A stack

Name	Description	Position
Push	Adding an element	Top of stack
Pop	Deleting an element	Top of stack

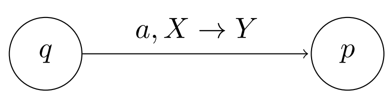
The above table summarizes the operations that can be performed on a stack.

- The length of the stack is unbounded.
- Contents of the entire stack is not visible to the automaton at any instant. Only the top element is accessible.
- To get to an element the automaton has to “pop” its way down. Elements that are popped are forgotten.

11.1.2 Informal Description of a Pushdown Automaton



- We allow the pushdown automaton to have a potentially different alphabet for its stack. Usually Γ denotes the stack alphabet.
- In case of a pushdown automaton, the transition function depends on,
  (i) the current state,
  (ii) the currently read input symbol (can be ϵ), and
  (iii) the currently read stack symbol at the top of the stack (again it can be ϵ).
- Each transition is of the form
  $$(q, a, X) \longrightarrow (p, Y).$$
  This is represented as follows:



  In this case, the automaton,
  (i) reads the input bit a (can be ϵ),
  (ii) moves from state q to state p,
  (iii) replaces the symbol X at the top of the stack with the symbol Y (both can be ϵ).

  Observe that if $X = \epsilon$ then it corresponds to pushing Y on the stack and if $Y = \epsilon$ then it corresponds to popping X from the stack.
- Since a pushdown automaton is non-deterministic, multiple possibilities can exist.

11.1.3 Formal Definition

Definition 11.1.1. A *pushdown automaton* (PDA in short) is a 6-tuple $(Q, \Sigma, \Gamma, \delta, q_0, F)$ where, Q is the finite set of states, Σ and Γ are the input and stack alphabet respectively, q_0 is the start state, F is the set of accept states, and
$$\delta : Q \times \Sigma_\epsilon \times \Gamma_\epsilon \longrightarrow 2^{Q \times \Gamma_\epsilon}$$
is the transition function of the automaton.

- A PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$ accepts a string $w \in \Sigma^*$ if
  - there exists $a_1, a_2, \dots a_m \in \Sigma_\epsilon$,
  - there exists states $r_0, r_1, \dots r_m \in Q$, and
  - there exists strings $s_0, s_1, \dots s_m \in \Gamma^*$,

  such that the following is true

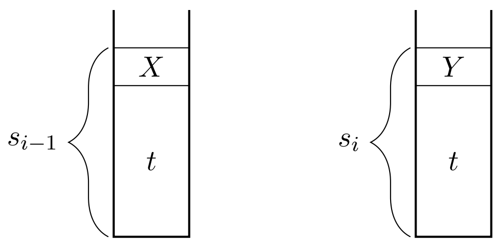
  (i) $w = a_1 a_2 \dots a_m$ (input is concatenation of the symbols being read)

  (ii) $r_0 = q_0$ and $s_0 = \epsilon$, (initial condition)

  (iii) $\forall i \in \{1, 2, \dots m\}$, if $(r_i, Y) \in \delta(r_{i-1}, a_i, X)$
  then $s_{i-1} = Xt$ and $s_i = Yt$, for some $X, Y \in \Gamma_\epsilon$ and $t \in \Gamma^*$. (transition condition)

  (iv) $r_m \in F$ (acceptance condition)

  The figure below shows how the stack changes in one transition step.



Before ith step After ith step

- The language of M,

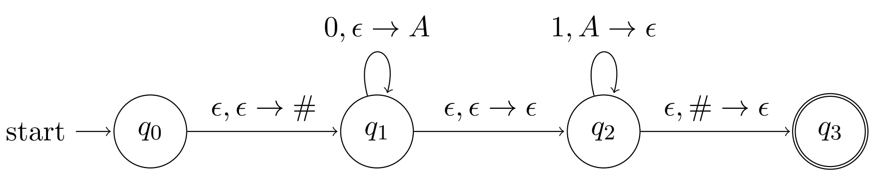
$$L(M) = \{w \in \Sigma^* \mid M \text{ accepts } w\}.$$

11.1.4 Example of a PDA

Consider the non-regular language

$$L = \{0^n 1^n \mid n \geq 0\}.$$

Idea: Push a symbol (say 'A') when reading a 0 from the input and pop the symbol when reading a 1. To check if the stack becomes empty exactly after all the 1's have been read, we initially push a 2nd symbol (say '#') on the stack without reading any input bit. Now after the 1's have been read if the stack contains the symbol # at its top then we accept the input else not. Below is the state diagram of the PDA for L.



We will now see the computation of some strings on this automaton.

Input string: $0^2 1^2$

Symbol read	Transition used	Current state	Stack contents
–	–	q_0	ϵ
ϵ	$(q_0, \epsilon, \epsilon) \longrightarrow (q_1, \#)$	q_1	$\#$
0	$(q_1, 0, \epsilon) \longrightarrow (q_1, A)$	q_1	$A\#$
0	$(q_1, 0, \epsilon) \longrightarrow (q_1, A)$	q_1	$AA\#$
ϵ	$(q_1, \epsilon, \epsilon) \longrightarrow (q_2, \epsilon)$	q_2	$AA\#$
1	$(q_2, 1, A) \longrightarrow (q_2, \epsilon)$	q_2	$A\#$
1	$(q_2, 1, A) \longrightarrow (q_2, \epsilon)$	q_2	$\#$
ϵ	$(q_2, \epsilon, \#) \longrightarrow (q_3, \epsilon)$	$\mathbf{q_3}$	ϵ

The input is accepted since the final state is an accept state and the input is completely read.

Input string: $0^2 1$

Symbol read	Transition used	Current state	Stack contents
–	–	q_0	ϵ
ϵ	$(q_0, \epsilon, \epsilon) \longrightarrow (q_1, \#)$	q_1	$\#$
0	$(q_1, 0, \epsilon) \longrightarrow (q_1, A)$	q_1	$A\#$
0	$(q_1, 0, \epsilon) \longrightarrow (q_1, A)$	q_1	$AA\#$
ϵ	$(q_1, \epsilon, \epsilon) \longrightarrow (q_2, \epsilon)$	q_2	$AA\#$
1	$(q_2, 1, A) \longrightarrow (q_2, \epsilon)$	q_2	$A\#$

Observe that no more transitions can be applied at this point and the current state of the automaton is not an accept state. Hence the input is not accepted.

Input string: 01^2

Symbol read	Transition used	Current state	Stack contents
–	–	q_0	ϵ
ϵ	$(q_0, \epsilon, \epsilon) \longrightarrow (q_1, \#)$	q_1	$\#$
0	$(q_1, 0, \epsilon) \longrightarrow (q_1, A)$	q_1	$A\#$
ϵ	$(q_1, \epsilon, \epsilon) \longrightarrow (q_2, \epsilon)$	q_2	$A\#$
1	$(q_2, 1, A) \longrightarrow (q_2, \epsilon)$	q_2	$\#$
ϵ	$(q_2, \epsilon, \#) \longrightarrow (q_3, \epsilon)$	q_3	ϵ

In this case although the automaton has reached an accept state, its input will not be accepted since the input is not completely read.

Note 7. Recall that in a non-deterministic automaton (finite or pushdown) an input is accepted if there is *some* sequence of transitions (also known as *computation path*) that leads to acceptance. Conversely an input is not accepted if *all* computation paths lead to rejection. In the above example, for each of the 2nd and 3rd strings, only one non-accepting computation path is shown. You can work out the other computation paths and verify that all of them lead to rejection.

Exercise 21. Construct PDA for the following languages

(i) $L_1 = \{w \in \{0,1\}^* \mid \#_0(w) = \#_1(w)\}$

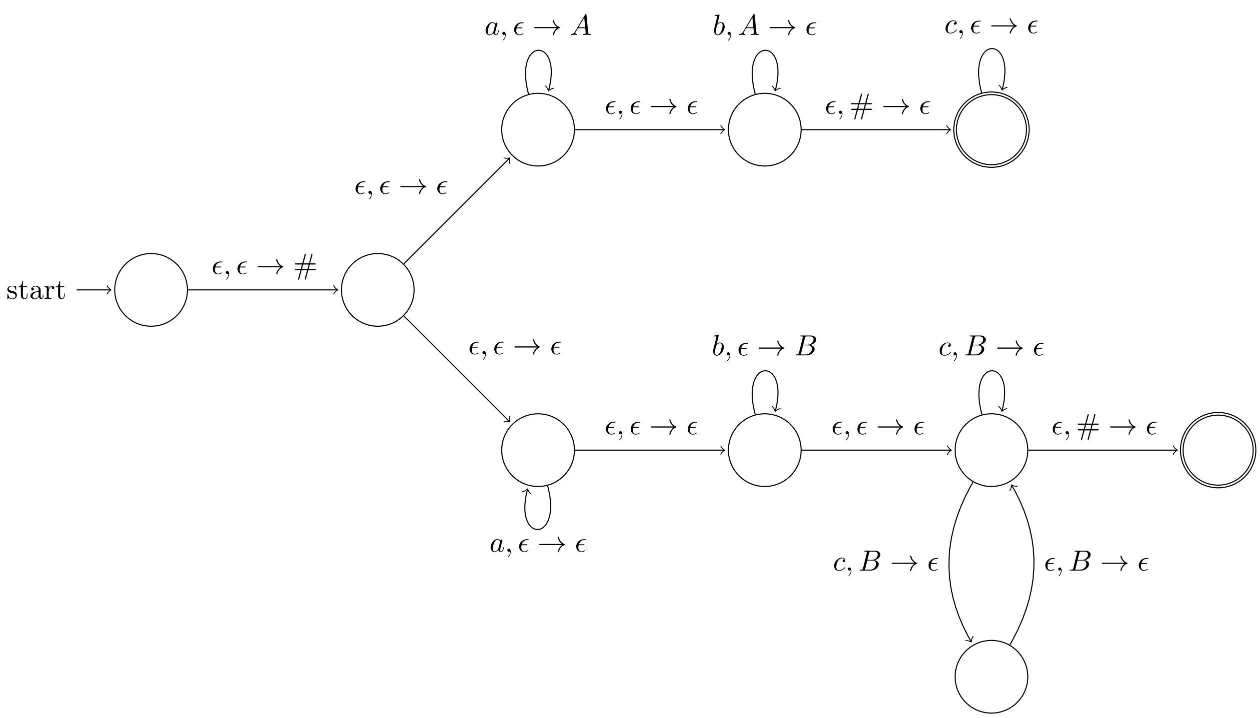
(ii) $L_2 = \{0^{2n} 1^{3n} \mid n \geq 0\}$

11.1.5 More Examples

1. Consider the language
$$L = \{a^i b^j c^k \mid i = j \text{ or } k \leq j \leq 2k\}.$$

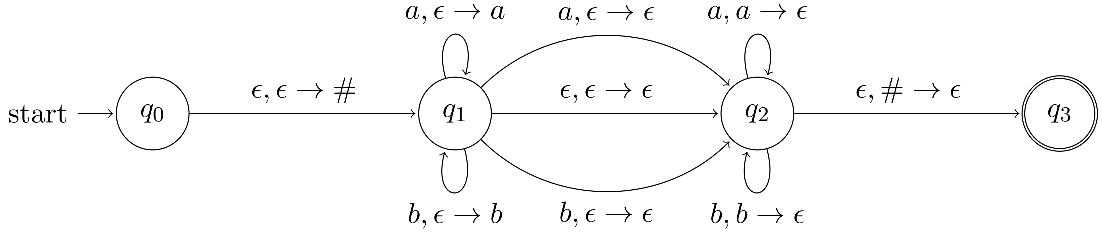
   We will use nondeterminism to decide which of the two conditions we are going to check and then proceed accordingly.

   To check whether $k \leq j \leq 2k$, we push a symbol to the stack for every b that we read and then for every c that we read, we again use non-determinism to pop either one or two of that symbol.



2. Another example
$$\text{PALINDROMES} = \{w = \text{rev}(w) \mid w \in \{a, b\}^*\}.$$

   We first push first half of the string in the stack and then compare the second half by popping symbols one at a time. But how do we figure out the middle position of the string? Use nondeterminism!



State q_1 pushes symbols onto the stack as they are being read and state q_2 pops symbols out of the stack as they are being read. If a computation path makes a transition from q_1 to q_2

before reaching the middle point then the stack will become empty before the string is read completely. On the other hand if a computation path makes a transition from q_1 to q_2 after reaching the middle point then the stack will contain some symbols above # even after the string is completely read.

11.1.6 CFGs and PDAs are equivalent

Theorem 19. *L is a CFL if and only if there exists a PDA M such that $L = L(M)$.*

Proof Idea. To show that if L is a CFL then L is accepted by some PDA start with a CFG and use the stack of the PDA to store the intermediate strings generated in each step of a derivation of a string. Terminals in the stack are "matched off" with the input of the PDA until a variable is reached. At this point, the PDA uses its non-determinism to replace the variable with one of its substitution rules and push it onto the stack. This process is carried on until the stack becomes empty and all the input symbols have been matched off.
For the other direction, we construct a CFG from a PDA. Without loss of generality assume that the PDA (a) has a single accept state, (b) empties the stack before accepting its input and (c) each transition either does a *push* operation or a *pop* operation but not both. The idea is to have variables of the form A_{pq} for each pair of states p, q. A_{pq} will generate all strings that takes the PDA from state p on an empty stack to state q on an empty stack (of course possibly popping/pushing some symbols at intermediate steps). A_{pq} is then defined inductively by dividing into two cases:

(i) *The first pushed symbol is same as the last popped symbol out of the stack.* Let a and b be the input symbols read at the first and last steps respectively, let r be the states following p and s be the state preceding q. Then we add the rule

$$A_{pq} \longrightarrow aA_{rs}b.$$

(ii) *The first pushed symbol is different from the last popped symbol.* Let r be an intermediate state when the stack becomes empty. In this case we add the rule

$$A_{pq} \longrightarrow A_{pr}A_{rq}.$$

□

Exercise 22 (Reading Exercise)**.** Read the complete proof from your textbook.

Exercise 23. Construct PDA for the following languages

(i) $L_1 = \{a^i b^j c^k \mid j \leq i + k \leq 2j\}$

(ii) $L_2 = \{a^i b^j \mid i \neq j\}$

(iii) $L_3 = L(a^*b^*c^*) \setminus \{a^n b^n c^n \mid n \geq 0\}$

(iv) $L_4 = \overline{L}$, where $L = \{ww \mid w \in \{a, b\}^*\}$